复旦数学资料汇编
数值代数与优化

来源：香蕉空间 Solution 资料库

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第一章　数值代数与优化

1.1　习题: 数值代数与优化

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1.2　24-25 学年第二学期第一周习题

Problem 1. [10 pt] For $x \in \mathbb{R}^m$ and $A \in \mathbb{R}^{m\times n}$, verify the following inequalities and give examples of a nonzero vector or matrix for which equality is achieved.

(a) $\|x\|_\infty \le \|x\|_2 \le \sqrt{m}\|x\|_\infty$

(b) $\frac{1}{\sqrt{n}}\|A\|_\infty \le \|A\|_2 \le \sqrt{m}\|A\|_\infty$

(c) Show equivalence between induced matrix norms $\|\cdot\|_p$ and $\|\cdot\|_q$ for $1 \le p < q \le \infty$

Problem 2. [10 pt] For each of the following statement, find an example matrix.

(a) a matrix whose singular values are the same as its eigenvalues;

(b) a matrix whose singular values are the same as the absolute values of its eigenvalues (the matrix has at least one negative eigenvalue);

(c) a matrix whose singular values are not the same as the absolute values of its eigenvalues;

Theorem I. Let $A \in \mathbb{R}^{m\times n}$ with SVD $\{\sigma_i, u_i, v_i\}_{i=1}^{\min(m,n)}$. For $0 \le r \le \min(m,n)$, the matrix A_r as defined,

$$A_r = \sum_{i=1}^{r} u_i \sigma_i v_i^*$$

satisfies

$$\|A - A_r\|_{\mathrm{F}} = \inf_{\substack{B\in\mathbb{C}^{m\times n} \\ \mathrm{rank}(B)\le r}} \|A - B\|_{\mathrm{F}}.$$

Problem 3. [20 pt] Prove the Theorem I.

Problem 4. [15 pt] For nonzero projector $P \in \mathbb{C}^{m\times m}$, show that $\|P\|_2 \ge 1$ with equality iff P is an orthogonal projector.

Problem 5. [20 pt] Implement a matrix-matrix multiplication algorithm in C++.

Problem 6. [10 pt] Call “dgemm” function in LAPACK for a matrix-matrix multiplication in C++.

Problem 7. [10 pt] Apply your code in Problem 5 and Problem 6 to two random matrices of size 5000×5000 and compare their runtime.

1.3 Homework 1

Due Mar 22, 2023

10 pt If u and v are m-vectors, the matrix $A = I - uv^*$ is known as a rank-one perturbation of the identity. Show that if A is nonsingular, A^{-1} is also a rank-one perturbation of the identity. For what u and v is A singular? If it is singular, what is its null space?

10 pt For $x \in \mathbb{R}^m$ and $A \in \mathbb{R}^{m\times n}$, verify the following inequalities and give examples of a nonzero vector or matrix for which equality is achieved.

- (a) $\|x\|_\infty \le \|x\|_2 \le \sqrt{m}\,\|x\|_\infty$;
- (b) $\frac{1}{\sqrt{n}}\,\|A\|_\infty \le \|A\|_2 \le \sqrt{m}\,\|A\|_\infty$;
- (c) Show the equivalence between induced matrix norms $\|\cdot\|_p$ and $\|\cdot\|_q$ for $1 \le p < q \le \infty$.

10 pt For each of the following statement, find an example matrix.

- (a) a matrix whose singular values are the same as its eigenvalues;
- (b) a matrix whose singular values are the same as the absolute values of its eigenvalues (the matrix has at least one negative eigenvalue);
- (c) a matrix whose singular values are not the same as the absolute values of its eigenvalues;

10 pt Complete the proof for the uniqueness of the singular value decomposition.

Theorem I. Given a matrix $A \in \mathbb{R}^{m\times n}$. The singular values of A are denoted as $\{\sigma_i\}_{i=1}^{\min(m,n)}$ with their corresponding left and right singular vectors being $\{u_i\}_{i=1}^{\min(m,n)}$ and $\{v_i\}_{i=1}^{\min(m,n)}$. For any r with $0 \le r \le \min(m,n)$, the matrix A_r as defined,

$$A_r = \sum_{i=1}^{r} u_i \sigma_i v_i^*,$$

satisfies

$$\|A - A_r\|_{\mathrm{F}} = \inf_{\substack{B\in\mathbb{C}^{m\times n} \\ \mathrm{rank}(B)\le r}} \|A - B\|_{\mathrm{F}}\,.$$

20 pt Prove the Theorem I.

15 pt Let $P \in \mathbb{C}^{m\times m}$ be a nonzero projector. Show that $\|P\|_2 \ge 1$, with equality if and only if P is an orthogonal projector.

15 pt Implement Classical Gram-Schmidt (CGS) algorithm.

15 pt Implement Modified Gram-Schmidt (MGS) algorithm.

15 pt Generate a matrix with exponentially graded singular values. Apply your own CGS, and MGS to the matrix. Plot the computed $r_j j$ verses j in logy style.

1.4 Homework 2

Due May 26, 2023

20 pt (a) The Wine Quality Data Set (`https://archive-beta.ics.uci.edu/dataset/186/wine+quality`) contains two datasets related to red and white variants of the Portuguese "Vinho Verde" wine. The goal here is to model **white wine** quality based on physicochemical tests. Assume the

white wine quality linearly depends on the results of physicochemical tests. Solving a least square problem,

$$\min_{\beta} \|X\beta - y\|_2$$

establishes the relation between physicochemical tests and wine quality, where $X \in \mathbb{R}^{4898\times 11}$ is the input variables, $y \in \mathbb{R}^{4898}$ is the output variable, and $\beta \in \mathbb{R}^{11}$ is the unknown coefficient vector. Use the QR method (your modified Gram-Schmidt implementation) to solve the least square problem and show the coefficient vector β and the corresponding error $\|X\beta - y\|_2$.

(b) For the same dataset, we modify the least square problem,

$$\min_{\beta,c} \|(X\beta + c) - y\|_2 \,,$$

where c is a scalar. Adding a scalar to a vector means that the scalar is added to each entry of the vector. Use the QR method again (your modified Gram-Schmidt implementation) to solve the least square problem and show the coefficient vector β, the scalar c and the corresponding error $\|X\beta + c - y\|_2$.

15 pt Implement Householder QR algorithm and Implicit Q and Q^* application, i.e., calculating Qx and Q^*x.

15 pt Describe an algorithm for QR factorization based on Givens rotations. And show that the Givens-QR is asymptotically 50% more expensive than Householder QR.

20 pt Let $A \in \mathbb{R}^{m\times k}$, $B \in \mathbb{R}^{k\times n}$ and $C = AB$. Show that there exists δC such that,

$$\widetilde{C} = AB + \delta C,$$

and an upper bound for δC (element-wise or matrix norm upper bound, you could also show both).

Lemma A. Let P be an exact Householder transformation, and $\widehat{P}$ be the computed one. Then

$$\mathrm{fl}(\widehat{P}A) = P(A + E)$$

with

$$\|E\|_2 = \|A\|_2\, O(\epsilon_{\text{machine}}).$$

20 pt Prove Lemma A.

1.5 Homework 3

Due Mar 30, 2023

20 pt Implement a memory-efficient LU factorization with partial pivoting, i.e., store L and U together in a $m \times m$ matrix and permutation matrix P as a vector of size m.

10 pt Implement the solve function associated with the memory-efficient LU factorization with partial pivoting.

5 pt Apply your LU factorization code to the following matrices,

$$A = \begin{pmatrix} 1 & & & & & & 1 \\ -1 & 1 & & & & & 1 \\ -1 & -1 & 1 & & & & 1 \\ \vdots & \vdots & \vdots & \ddots & & \vdots & \\ -1 & -1 & -1 & \cdots & -1 & 1 & 1 \\ -1 & -1 & -1 & \cdots & -1 & -1 & 1 \end{pmatrix}$$

of size 8×8 and 100×100.

20 pt Given an $(m+n) \times (m+n)$ symmetric positive definite matrix A of block form,

$$A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix},$$

where A_{11} is of size $m \times m$ and A_{22} is of size $n \times n$.

(a) Find a matrix L such that

$$LAL^\top = \begin{pmatrix} A_{11} & \\ & A_{22} - A_{21}A_{11}^{-1}A_{12} \end{pmatrix}.$$

The bottom-right block is known as the *Schur complement* of A_{11} in A.

(b) Show that the Schur complement is the same as m steps of Cholesky factorization algorithm.

10 pt Let A be a nonsingular square matrix and let $A = QR$ and $A^*A = U^*U$ be QR and Cholesky factorizations, respectively, with the usual normalizations $r_{jj}, u_{jj} \geq 0$. Is it true or false that $R = U$?

20 pt Implement an LDLT factorization with pivoting. Verify the correctness of your implementation using a random matrix of size 10×10.

1.6 Homework 4

Due Apr 16, 2023

20 pt Implement a Householder reduction to Hessenberg form for Hermitian matrices. Your implementation should fully explore the Hermitian property. Verify the correctness of your implementation using a random matrix of size 10×10.

20 pt Let $A \in \mathbb{C}^{m \times m}$ be tridiagonal and hermitian, with all its sub- and super diagonal entries nonzero. Prove that the eigenvalues of A are distinct.

40 pt Implement a QR algorithm for the eigenvalue problem of hermitian matrices. Wilkinson shift should be adopted (shift calculated by the eigenvalues of the bottom right 2×2 submatrix). Your function should return both eigenvalues and eigenvectors of the input matrix.

10 pt Generate a random 10×10 Hermitian matrix and calculate the eigenvalue decomposition using both your own implementation and system default function (either ”eig” in Matlab or ”cheevd” in Lapack). Compare the two outputs and validate that your implementation gives correct eigenvalue decomposition of the input matrices.

30 pt Let A be a symmetric matrix, x be a unit vector, $\rho(x, A) = \frac{x^\top Ax}{x^\top x}$ be the Rayleigh quotient. Let $r = Ax - \rho(x, A)x$ and α be the eigenvalue of A closest to $\rho(x, A)$. Prove,

$$|\alpha - \rho(x, A)| \leq \frac{\|r\|_2^2}{\min_{\alpha_i \neq \alpha} |\alpha_j - \rho(x, A)|},$$

where α_js are eigenvalues of A.

1.7 Homework 5

Due Apr 27, 2023

10 pt Let $A \in \mathbb{R}^{n\times n}$ be a *general* square matrix and $x, b \in \mathbb{R}^n$ be vectors. Compute the gradient and Hessian of $f_1(x) = b^\top x$ and $f_2(x) = x^\top Ax$.

20 pt Let $A \in \mathbb{R}^{n\times n}$ be a *general* square matrix and $X, B \in \mathbb{R}^{n\times k}$ be two matrices. Compute the gradient of $f_1(X) = \text{tr}(B^\top X)$, $f_2(X) = \text{tr}(X^\top AX)$, and $f_3(X) = \text{tr}(BX^\top AXX^\top BX^\top)$.

20 pt Show that the following conditions are equivalent for $\mu > 0$.

(a) $g(x) = f(x) - \frac{\mu}{2}\|x\|^2$ is convex for all x;

(b) $(\nabla f(x) - \nabla f(y))^\top (x - y) \geq \mu \|x - y\|^2$ for all x, y;

(c) $f(\alpha x + (1-\alpha)y) \leq \alpha f(x) + (1-\alpha)f(y) - \frac{\alpha(1-\alpha)\mu}{2}\|x - y\|^2$ for all x, y and $\alpha \in [0, 1]$.

10 pt Suppose that f is a convex function. Show that the set of global minimizers of f is a convex set.

30 pt Consider the following minimization problem,

$$\min_{x\in\mathbb{R}^n} f(x) = \frac{1}{2}x^\top Ax - x^\top b,$$

where $A \in \mathbb{R}^{n\times n}$ is a positive definite matrix, and $b \in \mathbb{R}^n$ is a vector. We apply the steepest descent method with a fixed stepsize to address the minimization problem, i.e.,

$$x_{k+1} = x_k - \alpha\nabla f(x_k),$$

where α is the constant stepsize. Please find a range for α such that the steepest descent method converges. For a fixed stepsize in the range, please prove that x_k converges to $x^* = A^{-1}b$ linearly.

1.8 Homework 6

Due May 7, 2022

30 pt Given a particular A and b. Suppose a breakdown of the Arnoldi iteration appear, i.e., $h_{k+1,k} = 0$.

(a) Show that K_k is an invariant subspace of A, i.e., $AK_k \subset K_k$.

(b) Show that $K_k = K_{k+1}$.

(c) Show that each eigenvalue of $Q_k^\top AQ_k$ is an eigenvalue of A.

(d) Show that if A is nonsingular, then the solution x to the system of equations $Ax = b$ lies in K_k.

10 pt Show that if one wishes to start GMRES with an arbitrary initial guess x_0, this can be accomplished by an easy modification of the right-hand side b.

15 pt Show that when applied to a quadratic function, with exact line searches, the Hestenes-Stiefel formula reduce to the Fletcher-Reeves formula.

25 pt Implement a GMRES using Givens QR for H_k. Apply your implementation to a real symmetric matrix A of size 805 (with eigenvalues being 1.00,1.01,1.02,...,8.99,9.00, and also 10,12,16,24). Check the number of iterations to reduce the initial error $\|e_0\|$ by a factor of 10^6.

10 pt Implement a GMRES with restart. Apply GMRES(20) to the symmetric matrix in Problem 4.

1.9 Homework 7

Due May 18, 2023

10 pt Show that if f is strongly convex, then we have $s_k^\top y_k \geq 0$ holds for any vectors x_k and x_{k+1}.

10 pt Using the updating equation of H_k in BFGS, derive the updating equation of B_k.

10 pt Show that the determinant of B_{k+1} admits,

$$\det(B_{k+1}) = \det(B_k)\frac{y_k^\top s_k}{s_k^\top B_k s_k}.$$

for B_k defined as in previous problem.

30 pt Self-study Limited-memory BFGS method (L-BFGS) and implement the method with a backtracking linesearch.

10 pt Apply your L-BFGS to an objective function $f(x) = \frac{1}{2}x^\top Ax - x^\top b$ with SPD A and vector b randomly generated. Try a few different m (number of columns) and plot the convergence curves for various m on the same figure.

0 pt Think about the connection between L-BFGS and CG method on quadratic polynomials as above.

1.10 Homework 8

Due Jun 4, 2023

20 pt Given an constraint optimization problem,

$$\min_{x\in\Omega} f(x),$$

where $f(x)$ is a convex function and the feasible set Ω is also convex. Show that all local solutions are also global solutions. In addition, the set of global solutions is convex.

15 pt Let $v : \mathbb{R}^n \to \mathbb{R}^m$ be a smooth vector function and consider the unconstrained nonsmooth optimization problem,

$$\min_{x\in\mathbb{R}^n} \max_{i=1,2,\ldots,m} |v_i(x)|.$$

Reformulate this problem as a smooth constrained optimization problem.

15 pt Prove that when the KKT conditions and the LICQ are satisfied at a point x^*, the Lagrange multiplier λ^* is unique.

20 pt Suppose that all constraints $c_i(x)$ are linear functions. Show that $T_\Omega(x) = \mathcal{F}(x)$ for all x in feasible set.

20 pt Prove that if the linear programming has solutions, then at least one such solution is a basic optimal point.

0 pt Consider the subproblem in trust-region method,

$$\min_{\substack{p\in\mathbb{R}^n \\ \|p\|\le\Delta}} f + g^\top p + \frac{1}{2}p^\top Bp.$$

p^* is a global solution of the subproblem if and only if p^* is feasible and there is a scalar $\lambda \ge 0$ such that the following conditions are satisfied,

$$(B+\lambda I)p^* = -g,$$

$$\lambda(\Delta - \|p^*\|) = 0,$$

$(B+\lambda I)$ is positive semidefinite.

[Hint: Using KKT conditions and Problem 1]

1.11　试卷: 数值代数与优化

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1.12　26 春期末考

本次考试两大题除最后一问外完全开卷, 可自由使用 ai, 需在 3 小时内完成并上传代码. 两道大题的最后一问是另发试卷, 在最后一小时内完成, 完全开放任何纸质资料. 总时长 4 小时.

数据集下载: `https://pan.baidu.com/s/1Twv-LoniVcjVI1fAYztqew?pwd=antc`

一、 Given a sparse matrix $A \in \mathbb{R}^{m\times n}$ and a vector $b \in \mathbb{R}^m$. We look for a vector x that minimizes the norm

$$\|Ax - b\|_2.$$

The matrix A and the vector b are given in the dataset.

(1) (10 pt) Find a property, e.g., sparsity, low-rank, etc., of the given matrices and vectors in the dataset. Based on this property, propose three potential methods to solve the problem.
[Mention the method names if they were discussed in class, or propose a short pseudocode and explain why the methods are suitable.]

(2) (15 pt) Design an algorithm to address the problem with matrices and vectors in the given dataset.

(3) (5 pt) Discuss the requirements of A and b such that your designed algorithm can be applied.

(4) (10 pt) Write the value of x_3 for each matrix-vector pair in the dataset.
[Keep 6 decimal digits in your answer.]

(5) (10 pt) (This problem was answered under closed-book conditions.) Given matrix $L \in \mathbb{R}^{p\times n}$, and a parameter $\lambda > 0$, consider the regularized least-squares problem

$$\min \|Ax - b\|_2^2 + \lambda\|Lx\|_2^2$$

Give a necessary and sufficient condition such that the above minimization problem has a unique minimizer.

二、 Consider the following constrained optimization problem:

$$\min_{\substack{\mathrm{ReLU}(c_i^\top x+\beta_i)=0\\ i=K+1,\dots,M}} \sum_{i=1}^{K} w_i \,\mathrm{ReLU}(c_i^\top x+\beta_i).$$

In the above problem, the ReLU function is defined as

$$\mathrm{ReLU}(z)=\max(0,z).$$

(1) (10 pt) Write a KKT-like condition for the above system. More precisely, please write the necessary conditions for the optimality of the above optimization problem.
[If the above problem cannot be answered directly, you could replace ReLU by ReLU^2 and then write down the KKT conditions.]

(2) (10 pt) Design and implement an algorithm to find a feasible point for the optimization problem with given data.
[The algorithm could adopt the property of the given dataset and be adapted to your following optimization solver.]

(3) (10 pt) Design and implement an algorithm to solve the optimization problem with given data.

(4) (10 pt) Apply the designed algorithm to the given matrices. Write the minimum objective values.
[Keep 6 decimal digits in your answer.]

(5) (10 pt) (This problem was answered under closed-book conditions.) Solve this problem by hand

$$\min_{\substack{\mathrm{ReLU}(x_1+2x_2-x_3+4)=0\\ \mathrm{ReLU}(-2x_1+x_2+x_3+3)=0\\ \mathrm{ReLU}(x_1-x_2+3x_3+-6)=0\\ \mathrm{ReLU}(-x_1+4x_2+x_3+7)=0}} \mathrm{ReLU}(2x_1-3x_2-4x_3-2)$$

1.13　25 春期末考

第一题, 除矩阵向量运算外, 任何作为解答的算法需要自行实现, 不能直接调用矩阵分解, 线性方程组求解等库函数. 例如, 若将 LU 分解作为解答, 需要自行实现 LU 分解算法, 不得直接调用 MATLAB 自带 [L,U]=lu(A) 函数.

第二题, 允许调用矩阵分解等库函数, 但不允许直接调用优化求解器.

一、 Given a symmetric matrix $A\in\mathbb{R}^{n\times n}$. We look for a unitary matrix W such that,

$$W=\begin{pmatrix} A & * \\ * & * \end{pmatrix},$$

where $*$ denotes an unspecified block of the matrix. The matrix given in the dataset is A^2.

(1) (10 pt) Give an if-and-only-if condition on A such that the unitary matrix W exists.

(2) (10 pt) Assume A could be embedded into a unitary matrix W, where the size of W is a power of 2, i.e., $W\in\mathbb{R}^{2^k\times 2^k}$ for some integer k. Design a unitary matrix W.

(3) (20 pt) Design an algorithm taking A^2 as the input and compute the application of W on a given vector.
[Your algorithm should include factorization of A. The algorithm could be tailored to the specific structure of A^2 in the dataset.]

(4) (10 pt) Let n be the size of A. Embed A into W of size $2n$. Write the value of

$$\begin{pmatrix} 1 & 1 & 0 & \cdots & 0 \end{pmatrix} W(e_1 + e_{n+1}),$$

for each A^2 in the dataset, where $e_i \in \mathbb{R}^{2n}$ is a zero column vector with 1 at the i-th position.

二、 Given two nonsingular matrices $A \in \mathbb{R}^{m\times k}$ and $B \in \mathbb{R}^{k\times n}$ with $k < \min(m, n)$, a weight vector $c \in \mathbb{R}^n$, and a vector $b \in \mathbb{R}^m$. Consider the following linear program,

$$\min_{\substack{ABx=b \\ x\geq 0}} c^\top x.$$

(1) (10 pt) Write the KKT condition for above system.

(2) (20 pt) Design and implement an algorithm to solve the linear program.
[The algorithm could be tailored to the specific structure of matrices in the dataset.]

(3) (5 pt) Based on the given datasets, analyze the computational complexity, and give a complexity upper bound for your designed algorithm.

(4) (15 pt) Apply the designed algorithm to the given matrices. Write the minimum objective values.
[Keep 6 decimal digits in your answer.]